Raj Tatavarthi

Use Case UC1: Create and Customize

Primary Actor: User

Stakeholder and Interests

- User : Wants to create a personalized avatar that represents their identity
- System: Wants to store and render the avatar data

Preconditions

- User has created an account and is logged in

Success Guarantee
- The user creates their own avatar and system saves it and displays it throughout the app

Main Success scenario

- 1. User opens the Avatar Customization screen
- 2. System displays base avatar options like body, skin tone, hair, and etc
- 3. User clicks on one of the options like skintone or hairstyles
- 4. The system applies what the user selected onto to the avatar
- 5. User selects all the option from each category
- 6. The system renders the avatar
- 7. User selects “Save Avatar”
- 8 System saves the avatar and adds it to the profile

Extension
- User selects a premium locked option
- Saves the information when their network error

Special Requirements
- Avatar preview must render with a certain time

Technology and Data Variation
- Avatar data stored in the user profile

Use Case UC2: apply theme to Interface

Primary Actor: User

Stakeholder and Interest

- User: Wants to personalize the app's visual appearance
- System: Wants to apply the theme consistently across all screens

Preconditions
- User is logged in.
- User has navigated to Settings > Themes

Success Guarantee:
- The selected theme is applied globally across all app screens

Main Success Senario:
- 1. User opens the Appearance settings screen.
- 2. System displays available themes like Light, Dark, and Ocean Blue
- 3. User taps a theme to preview it
- 4. System renders the theme selected by the user
- 5. User taps "Apply Theme
- 6. System applies the theme globally to all screens
- 7. System applies the theme to the dashboard

Extension
- User previews multiple themes before choosing

Special Requirements
- Theme transitions must be smooth

Technology and Data Variation
- Theme stored as a theme ID string in the user profile

Use Case UC3: Customize Dashboard

Primary Actor: User

Stakeholder and Interest

- User: Wants to arrange and configure dashboard widgets
- System: Wants to render the user's custom layout

Preconditions
- User is logged in
- User has opened the Dashboard screen and activated Edit Mode

Success Guarantee
- The custom dashboard layout is saved

Main Success Senario

- 1. User taps the "Edit Dashboard" button.
- 2. System enters edit mode showing all available widgets with drag handles
- 3. User drags and repositions widgets to their preferred layout
- 4. System shows a live preview of the updated layout
- 5. User taps "Save Layout"
- 6. System saves the layout configuration to the user's profile
- 7. System exits edit mode and displays the updated dashboard

Extension

- System snaps the dragged widget to the nearest available open slot
- System restores the default widget layout and saves it

Special Requirements

- Dashboard must load in under 1 second with up to 10 active widgets

Technology and Data Variation

- Layout stored

Use Case UC4: Complete Onboarding Tutorial

Primary Actor: New User

Stakeholder and Interests

- New User: Wants to quickly learn the app's core features to start using it effectively
- System: Wants to guide users through key features

Preconditions

- User has just created a new account and is opening the app for the first time
- Tutorial has not been previously completed or dismissed

Success Guarantee

- User completes all tutorial steps and is directed to the main dashboard

Main Success Scenario

- 1. System detects first time user and autolaunches the onboarding tutorial
- 2. System displays a welcome screen with app name and a brief value summary
- 3.User taps "Get Started"
- 4. System walks user through tutorial
- 5. User taps "Start Using the App"

- 6. System navigates the user to the main dashboard

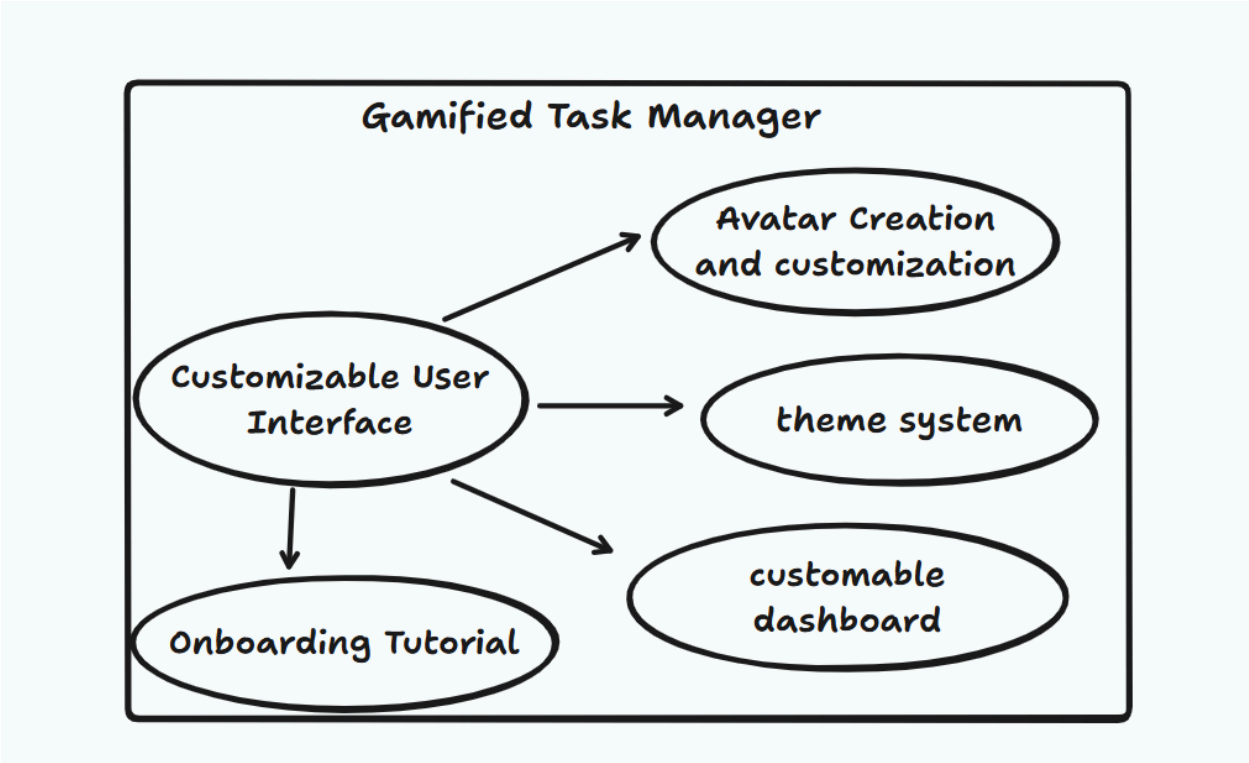
Extension
- At any step, user taps "Skip Tutorial"

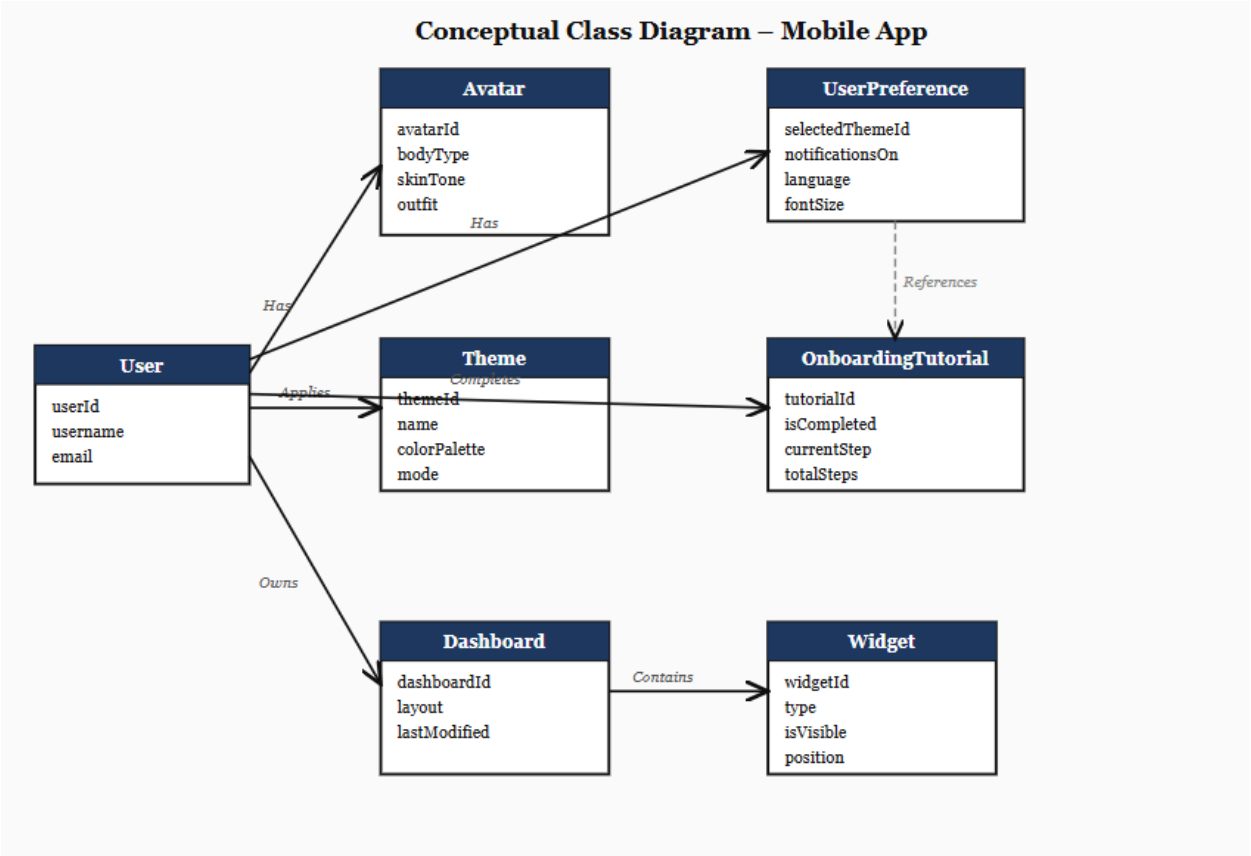
Special Requirement
- Tutorial must be completable offline

Technology and Data
- Tutorial screens implemented as a step-through model

NFR1: Responsive Design

- The system should work well on both phones and computers. The layout should automatically adjust to fit different screen sizes so users can easily access all features no matter what device they're using. The app should also load quickly, within 3 seconds, on any device.





Vivan

Use Case 01: Award XP on Task Completion

Primary Actor: User

Stakeholders and Interests:

- User: wants XP to feel "right"
- Product Team: wants XP rules that drive engagement
- System Admin: wants XP awards logged

Preconditions:

- User is authenticated
- Task exists and not completed
- XP rule configuration is active

Success guarantee (Postconditions):

- XP is added exactly once
- XP event is recorded
- Updated XP is visible in UI

Main success scenario:

1. User marks task as completed.
2. System verifies task eligibility.
3. System calculates XP.
4. System applies eligible multipliers if needed.
5. System records XP event.
6. System updates user's total XP.
7. System updates UI in real time.

Extensions:

- 2a. Task already completed, System denies XP and shows message.
- 3a. XP rule missing, Default XP applied and error logged.
- 5a. Database failure, XP not awarded user notified

Special requirements:

- XP award must be one time thing

Technology and data variation list:

- Web client
- Mobile client
- Offline sync mode

Use Case 02: Level Up via Progression Thresholds

Primary Actor: User

Stakeholders and Interests:

- User: wants satisfying progression pacing
- Product Team: wants adjustable threshold system

Preconditions:

- User has XP value stored
- Level thresholds configured

Success guarantee:

- User level reflects correct XP range
- Level-up event recorded
- UI reflects new level immediately

Main success scenario:

1. XP total changes.
2. System compares XP to threshold table/formula.
3. If threshold crossed, system updates level.
4. Level-up event recorded.
5. UI displays updated level and progress bar.

Extensions:

- 3a. Multiple levels crossed, All applied in single update.
- 2a. Threshold configuration updated, Level recalculated.

Special requirements:

- Level must be deterministically derived from XP.

Technology and data variation list:

- Threshold table stored in database
- Formula-based progression system

Use Case UC 03: Track Daily and Weekly Streaks

Primary Actor: User

Stakeholders and Interests:

- User: wants streak rewards without harsh punishment
- Product Team: wants configurable grace rule

Preconditions:

- User authenticated
- Time zone configured
- At least one eligible task completed

Success guarantee:

- Daily streak updated correctly
- Weekly streak updated correctly
- Streak state visible in UI

Main success scenario:

1. User completes an eligible task.
2. System checks if streak already counted today.
3. System updates daily streak.
4. System updates weekly streak.
5. System applies grace rule if applicable.
6. UI updates streak display.

Extensions:

- 2a. Multiple completions same day, Only first counts.
- 5a. Grace rule triggered, Streak preserved.

Special requirements:

- Streak calculation must be consistent across devices.

Technology and data variation list:

- Server-side time validation
- Client time display

Use Case UC 04: Apply Bonus Multipliers

Primary Actor: User

Stakeholders and Interests:

- User: wants noticeable but fair bonuses
- Product Team: wants capped multiplier system

Preconditions:

- Task completed

- Base XP calculated
- Multiplier rules configured

Success guarantee:

- Final XP calculated correctly
- Multiplier breakdown visible
- XP event logged

Main success scenario:

1. System calculates base XP.
2. System checks streak multiplier eligibility.
3. System checks early completion bonus eligibility.
4. System combines multipliers
5. Final XP awarded.
6. UI displays XP breakdown.

Extensions:

- 4a. Multiplier exceeds cap, Cap applied.
- 3a. Missing due date, Early bonus skipped.

Special requirements:

- A multiplier system must prevent abuse/exploit attempts.

Technology and data variation list:

- Configurable multiplier rules in database

Supplementary Specification

NFR 01: Real Time XP and Progress Updates

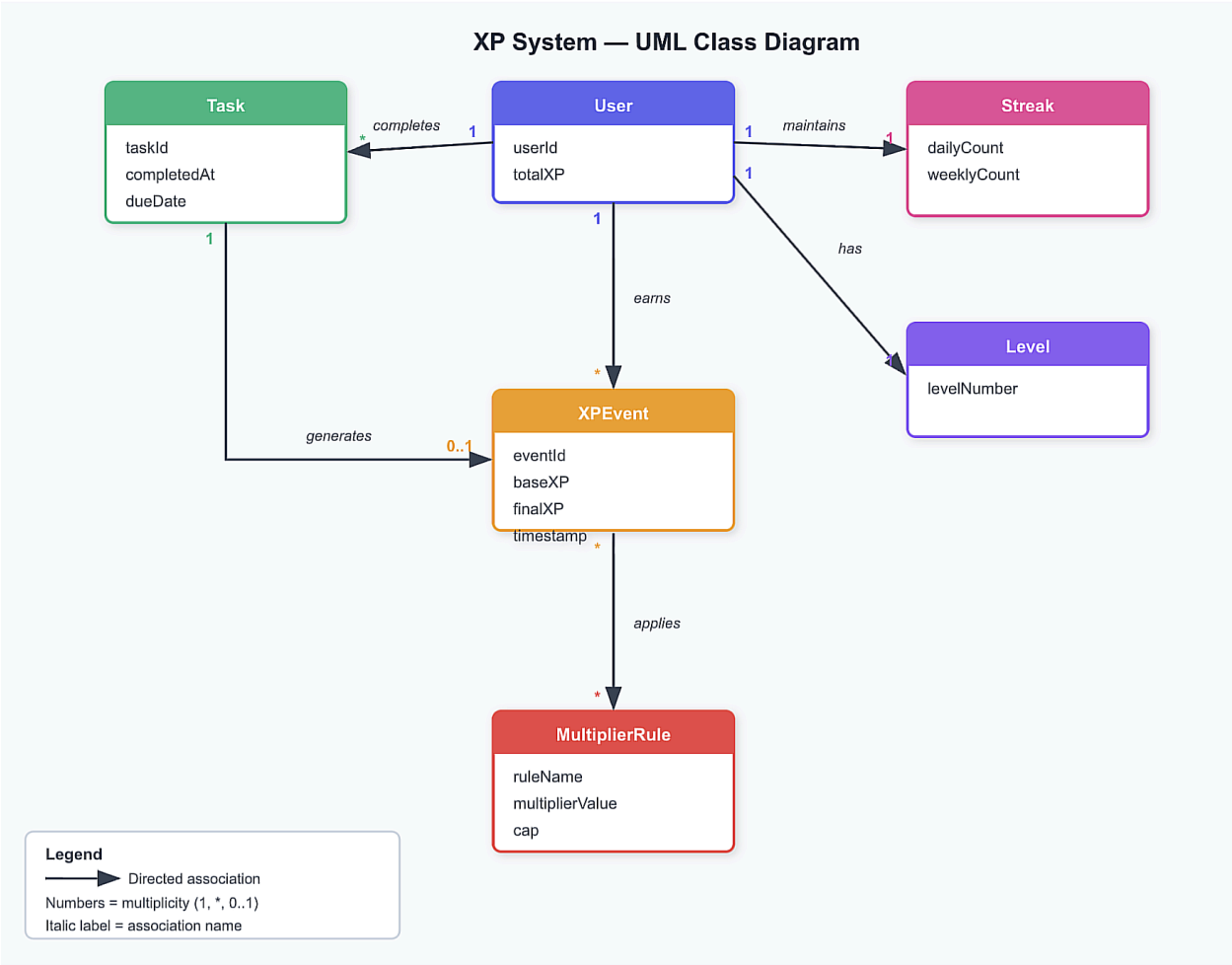
Category: Performance / User Experience

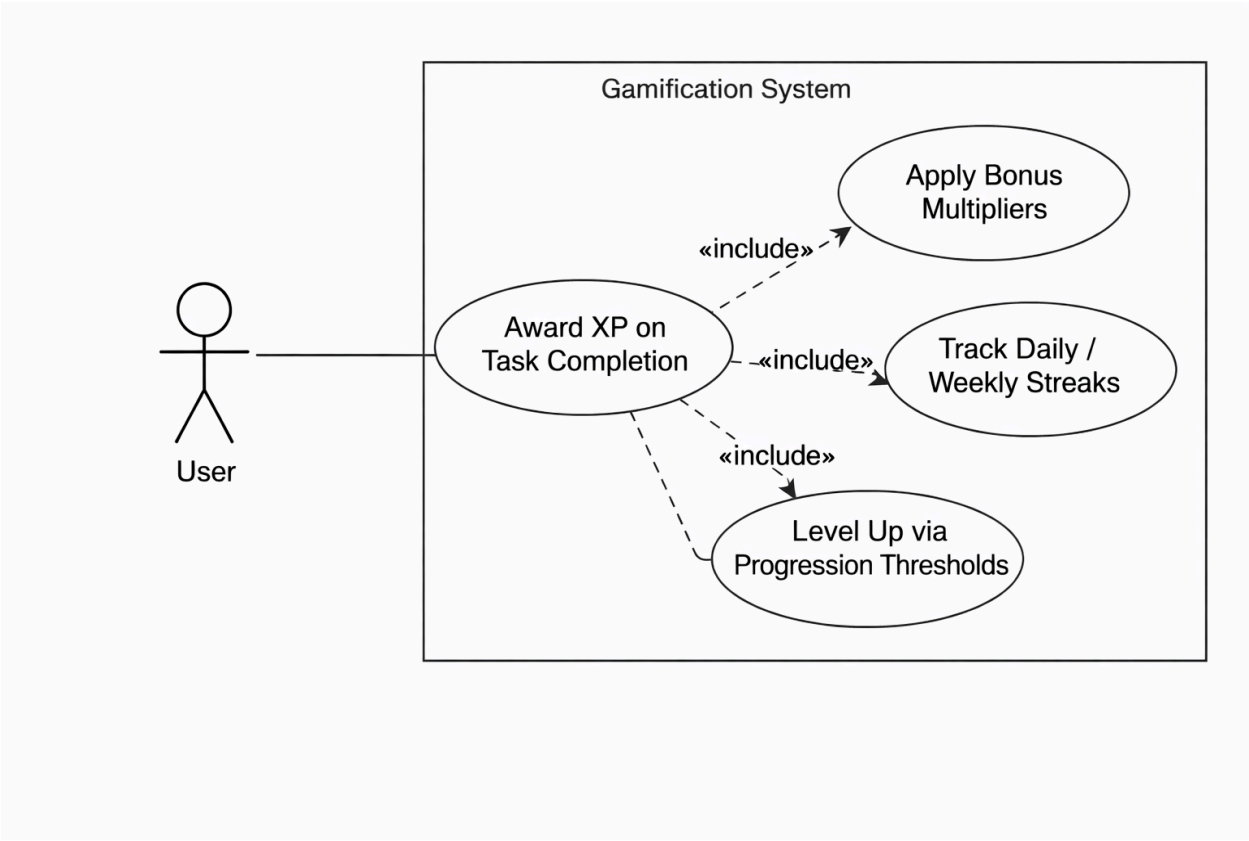
Requirement:
When a user's XP, level, or streak changes, the updated values must appear consistently across all UI surfaces immediately.

Explanation:
Progression feedback must feel instant to maintain engagement.

Acceptance Criteria:

- XP/level updates visible within 1 second.
- No inconsistent values between pages.
- Offline mode shows syncing state until confirmed.


XP System — UML Class Diagram
Task
taskId
completedAt
dueDate
User
userId
totalXP
Streak
dailyCount
weeklyCount
Level
levelNumber
XPEvent
eventId
baseXP
finalXP
timestamp
MultiplierRule
ruleName
multiplierValue
cap
completes
maintains
has
earns
generates
applies
1
*
0..1
Legend
Directed association
Numbers = multiplicity (1, *, 0..1)
Italic label = association name



Satwik

Use Case UC1: User Authentication and Profile Setup

Primary actor: User

Stakeholders and interests:

- User: wants secure, fast access to a personalized account.
- System Admin: wants authentication events logged and credentials stored securely
- Product Team: what is necessary is a smooth onboarding process that minimizes drop-off

Preconditions:

- The app is installed and reachable. No active sessions exist for this user.

Success and Guarantee:

- The user is authenticated and has an active session.
- A profile with username and default settings is created (or it is loaded if returning).
- The authentication event is logged.

Main Success Scenario:

1. The user opens the app and selects ‘Sign Up’.
2. The user enters email, username, and password.
3. The system validates that the email is not already registered and the password meets the naming requirements.
4. The system creates the account, hashed and stored the password.
5. The system creates a default user profile.
6. The system starts an authenticated session.
7. The user is directed to the home dashboard.

Extensions:

- 2a. User selects 'Log In' instead of Sign Up: System prompts for email and password, validates credentials, and starts a session. If credentials are incorrect, the system notifies the user and allows retry.

- 2b. User selects 'Continue with Google': System sends authentication to Google OAuth, then creates or loads the account.

- 3a. Email already registered: System notifies users and prompts them to log in or reset their password.
- 3b. Password does not meet requirements: System signals the error and prompts re-entry.

Special Requirements:

- Passwords must be hashed before storage. All communication must occur over HTTPS.

Technology and Data Variation List:

- 2a. Authentication may use email/password or OAuth (Google, Apple)

- 6a. Session token may be stored as a secure HTTP-only cookie or in device secure storage on mobile.

Use Case 2 UC2: Friends List Management

Primary Actor: User

Stakeholders and interests:

- User (requester): wants to add friends to unlock social features and compete.
- User (recipient): wants control over who can connect with them.
- Product Team: wants social connections to drive engagement and retention.

Preconditions:

- Both users are authenticated. The requester knows the recipient's username.

Success and Guarantee:

- A friend connection is established and reflected in both users' friends lists or an existing connection is removed/blocked as requested.

Main Success Scenario:

1. The user navigates to the Friends section.
2. The user searches for another user by username.
3. The system retrieves and displays the matching profile.
4. User selects 'Send Friend Request'.
5. The system records the pending request and notifies the recipient.
6. The recipient opens the notification and accepts the request.
7. The system adds both users to each other's friends lists.
8. The system confirms the new connection to both users.

Extensions:

- 2a. No user found for that username: System displays 'No results found' and prompts retry.
- 4a. A pending request already exists between these users: System displays 'Request Already Sent'.
- 4b. Users are already friends: System shows 'Already Friends' instead of the request button.
- 6a. Recipient declines the request: System removes the pending request and notifies the requester.
- 6b. Request not responded to within 7 days: System automatically expires the invitation.
- User selects 'Remove Friend': System prompts confirmation, then removes the connection from both lists.
- User selects 'Block User': System removes the connection and prevents the blocked user from sending future requests or viewing the user's profile.

Special Requirements:

Blocked users must be restricted from the blocking user's profile and social data immediately after the action.

Technology and Data Variation List:

- 2a. Search may be by username or email.
- 5a. Notifications may be delivered as app alerts or push notifications depending on device settings.

UC3: Use Case 3: View Leaderboard

Primary actor: User

Stakeholders and interests:

- User: wants to see accurate rankings against friends and global players to stay motivated.
- Product Team: wants leaderboards to reinforce daily engagement and task completion.
- Friends: want their XP scores reflected accurately and updated promptly.

Preconditions:

The user is authenticated. At least one user in the relevant group has XP from completed tasks.

Success and Guarantee:

The leaderboard is displayed with correct rankings for the selected scope and time period. The viewing user's rank is highlighted.

Main Success Scenario:

1. The user navigates to the Leaderboard section.
2. The system displays the leaderboard defaulting to 'Friends Only' and 'Weekly' view.
3. The system retrieves XP data for the user and all friends for the selected period.
4. The system ranks users by XP earned and displays rank, username, avatar, level, and XP.
5. The system highlights the viewing user's own row.
6. Users optionally change the time filter to Daily, Weekly, or All-Time.
7. The system refreshes rankings based on the new selection.

Extensions:

- 3a. No XP activity in selected period: System displays 'No activity yet' and encourages task completion.
- 6a. User switches scope to Global: System displays rankings across all app users instead of friends only.
- 3b. A friend's profile is set to private: System excludes that friend from the leaderboard.

Special Requirements:

Leaderboard rankings must reflect XP updates promptly after task completion to feel fair and responsive.

Technology and Data Variation List:

- 6a. Leaderboard may be sorted by XP earned, total tasks completed, or current streak depending on selected mode.
- 4a. Avatars display as default generated icons if a user has not set a custom avatar.

Use Case 4: 1 v 1 Task Completion

Primary Actor: User(challenger)

Stakeholders and interests:

- User (Challenger): wants to compete directly with a friend for motivation.
- User (Opponent): wants a fair, opt-in competition with clear rules.
- Product Team: wants challenges to drive task completion and social retention.

Preconditions:

Both users are authenticated and on each other's friends list. Both users have active accounts.

Success and Guarantee:

The challenge runs for its full duration, a winner is declared based on tasks completed, both users' stats are updated, and both are notified of the result.

Main Success Scenario:

1. The user navigates to the Friends section and selects a friend.
2. The user selects 'Challenge' and sets the parameters: duration and task goal.
3. The system sends a challenge invitation to the selected friend.
4. Friend receives the notification and accepts.
5. Both users complete tasks during the challenge period.
6. The system tracks task completions for both users.
7. At the end of the period, the system determines the winner based on tasks completed.
8. The system updates both users' stats and challenge history.
9. The system notifies both users of the result.

Extensions:

- 4a. Friend declines the challenge: System notifies the challenger and cancels the request.
- 4b. Friend does not respond within 24 hours: System automatically expires the invitation.
- 7a. Both users complete the same number of tasks: System declares a tie and rewards both users equally.

- 6a. One user's session disconnects mid-challenge: System resumes tracking when reconnected; completions made offline sync upon reconnection.

Special Requirements:

Challenge results must be finalized by the system and must not be modifiable by either user after the challenge period ends.

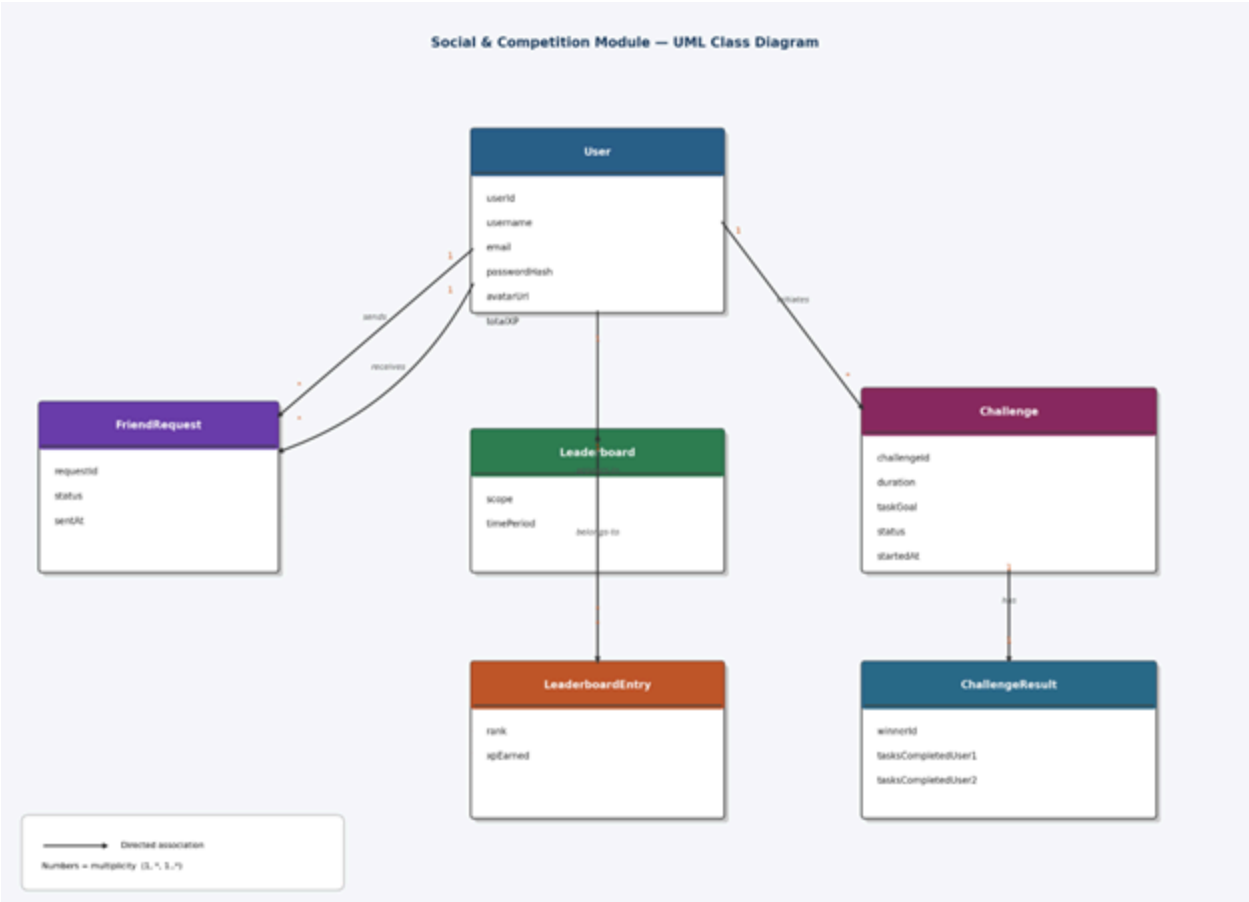
Technology and Data Variation List:

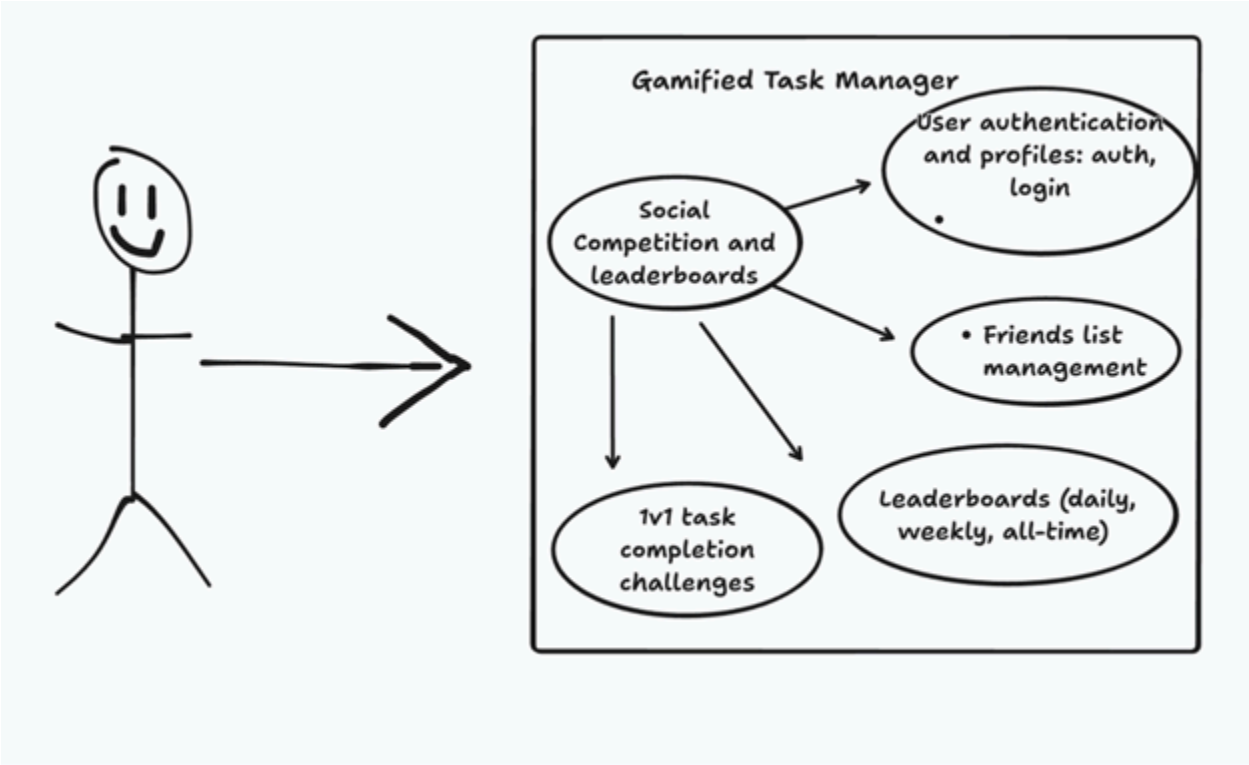
- 6a. Task completion tracking may occur via real-time sync or periodic polling depending on network conditions.
- 3a. Challenge invitations may be sent as an in-app notification or push notification.

Non-Functional Requirement: Security Authentication and Data Privacy

Requirement:
All user credentials must be stored using industry-standard hashing. All client-server communication must occur over HTTPS. User profile data, friends lists, challenge history, and leaderboard data must not be accessible to other users beyond what the application exposes on the interface to the public. Users must be able to remove friends, block users, and permanently delete their account and all personal data at any time.
Acceptance Criteria:

- Passwords are never stored in plaintext.
- All API endpoints require a valid authenticated session token.
- Blocked users cannot view the blocking user's profile, leaderboard entries, or send requests.
- Account deletion removes all personal data from the system within a reasonable time.


Social & Competition Module — UML Class Diagram
User
userId
username
email
passwordHash
avatarUrl
totalXP
sends
receives
initiates
FriendRequest
requestId
status
sentAt
Leaderboard
scope
timePeriod
belongs-to
Challenge
challengeId
duration
taskGoal
status
startedAt
has
LeaderboardEntry
rank
spEarned
ChallengeResult
winnerId
tasksCompletedUser1
tasksCompletedUser2
Directed association
Numbers = multiplicity (1, *, 1..*)


Gamified Task Manager
User authentication and profiles: auth, login
Social Competition and leaderboards
• Friends list management
1v1 task completion challenges
Leaderboards (daily, weekly, all-time)

Focus & Productivity Tools (Team)

This feature gives users tools that help them stay focused while working on tasks. It includes a focus timer similar to the Pomodoro technique, the ability to set daily goals, optional website blocking during focus sessions, and a page where users can review their past focus sessions.

Use Case UC21: Start a Focus Session (Pomodoro Timer)

Primary Actor
Registered User

Stakeholders & Interests

- User: wants a structured timer that helps them stay focused while working on a task and earn extra XP for completing focus sessions.

Preconditions

- User is logged in.
- At least one task exists in the user's task list.

Success Guarantee

- The focus session finishes successfully.
- Bonus XP is awarded.
- The session is saved in the user’s session history.

Main Success Scenario

1. User selects a task and chooses Start Focus Session.
2. System displays a countdown timer (default: 25 minutes work, 5 minute break).
3. Timer counts down while showing progress visually.
4. When the timer finishes, the system notifies the user and suggests taking a break.

5. After the break, the system asks the user if they want to start another session or mark the task as complete.

6. The system records the session and awards +10 XP.

Extensions

- 2a. User changes the session length (between 5 and 90 minutes). The system starts the session with the new time.

- 4a. User stops the timer early. The session is saved as partial and no XP is awarded.

Special Requirements

- Timer accuracy should be within ±1 second.

- Notification sound must be optional or dismissible.

Technology & Data Variation

- Session data is stored with start time, end time, task ID, and completion status.

Use Case UC22: Set and Track Daily Goal

Primary Actor
Registered User

Stakeholders & Interests

- User: wants to set a daily target for completed tasks to stay productive and feel motivated.

Preconditions

- User is logged in.

Success Guarantee

- Daily goal is saved and shown on the dashboard.

- When the goal is reached, bonus XP is awarded.

Main Success Scenario

1. User opens the Daily Goal Settings page.
2. User enters the number of tasks they want to complete for the day (for example, 5).
3. System saves the goal and displays a progress indicator on the dashboard.
4. Each time a task is completed, the progress indicator updates.
5. When the goal is reached, the system shows a celebration message and awards bonus XP.

Extensions

- 2a. User enters 0 tasks. The system shows a validation error.
- 5a. Goal has already been reached earlier in the day. The system displays a message but does not give additional XP.

Special Requirements

- The daily goal resets at midnight based on the user's time zone.

Technology & Data Variation

- The goal value is stored per user.
- Progress is calculated using the tasks completed that day.

Use Case UC23: Block Distracting Websites During Focus

Primary Actor
Registered User (Desktop / Browser Extension)

Stakeholders & Interests

- User: wants to avoid distractions by blocking certain websites during focus sessions.
- System: must enable blocking only while the focus session is active.

Preconditions

- User is logged in.
- A focus session is currently active.
- Browser permissions for blocking sites have been granted.

Success Guarantee

- Selected websites are blocked during the session.
- Blocking is removed automatically when the session ends.

Main Success Scenario

1. User opens Focus Settings and adds websites to the block list.
2. User starts a focus session.
3. System activates the blocking rules.
4. If the user attempts to open a blocked site, the system shows a message encouraging them to stay focused.
5. When the session ends, the system removes all blocking rules.

Extensions

- 1a. Required permissions are missing. The system asks the user to enable them first.
- 4a. User disables blocking during the session. The system records the override and removes the bonus XP.

Special Requirements

- Blocking must activate within 1 second after the session starts.
- No browsing data from blocked sites is stored.

Technology & Data Variation

- The block list is stored as a list of domain names for each user.

- Blocking is enforced through a browser extension.

Use Case UC24: Review Focus Session History

Primary Actor
Registered User

Stakeholders & Interests

- User: wants to review past focus sessions to see how productive they have been over time.

Preconditions

- The user is logged in.
- At least one focus session has been completed.

Success Guarantee

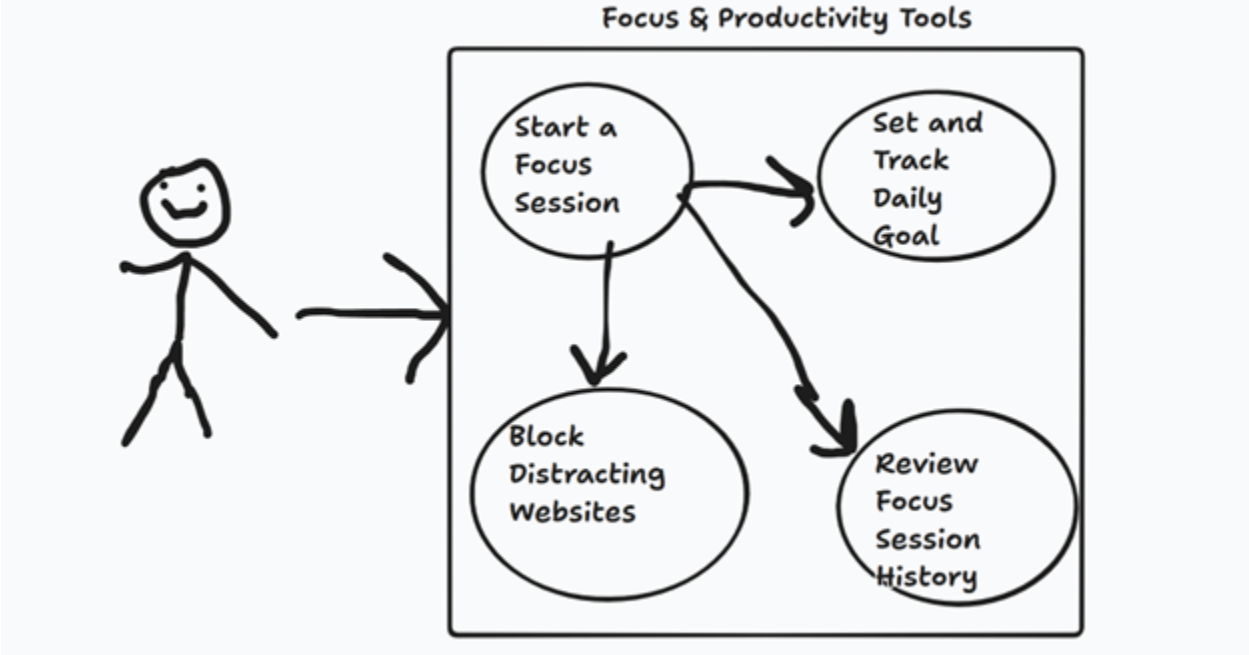
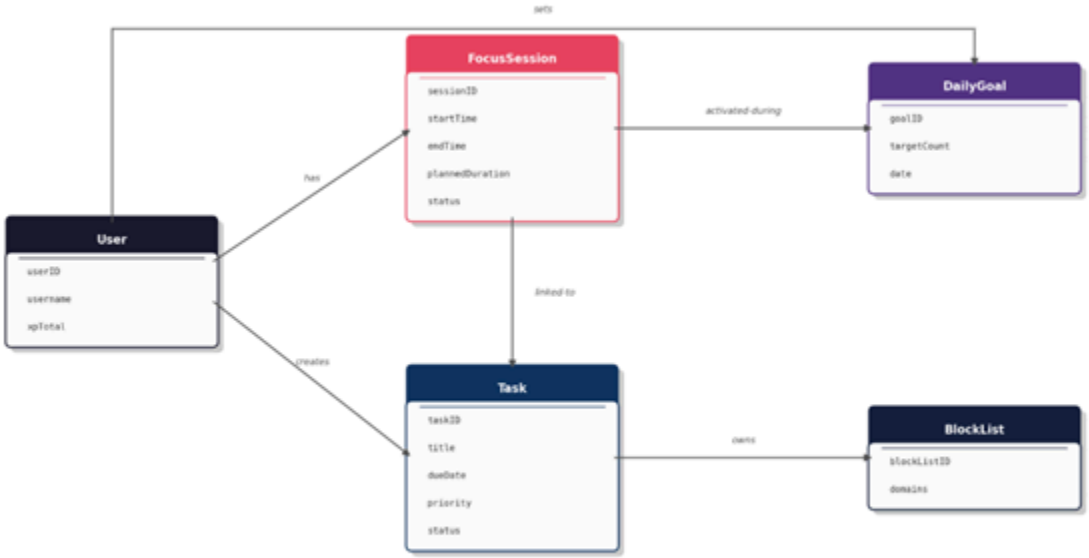
- The system displays a list of past sessions and summary statistics.

Main Success Scenario

1. User opens the Focus History page.
2. System retrieves the user's focus sessions.
3. Sessions are displayed with date, duration, linked task, and status (Complete or Partial).
4. A summary section shows total focus time (today, this week, and all time).
5. User can filter sessions by date or task category.

Extensions

2a. If no sessions exist yet, the system shows a message encouraging the user to start their first session.

Special Requirements

The page must load within 2 seconds when displaying up to 1000 sessions.